Numerical Sharpness of Global, Local, and Discrete Gaussian Fisher–Rao Convergence Rates

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Abstract

We numerically instantiate the adversarial constructions behind the sharp convergence rates for Gaussian Fisher–Rao gradient flow and its Riemannian retraction and KL/Bregman discretizations. Unlike earlier benchmarks on typical low-dimensional targets, this study is a lower-bound experiment: every target is selected to realize a specific obstruction. Deterministic quadrature and symmetry reduction give the fitted global exponents $T_{\mathrm{glob}} \sim \kappa_\star^{0.968}$ for the continuous flow and $N_{\mathrm{glob}} \sim \kappa^{2.001}$ for both fixed-step schemes. For the local spectral gap, tail fits against the claimed sharp scales give exponents 1.045 for $\log(e\kappa)$ in one dimension, 0.983 for $\sqrt{n\log(e\kappa)}$ on gamma shells, 0.923 for ridge saturation, and 0.957 for shell saturation against $\sqrt{\kappa}$. Nonlinear local trajectories recover the computed continuous and discrete linearized rates, and a single bump-train target displays the global-to-local change of slope and the benefit of increasing the stepsize after certified local entry.

1 Claims tested and experimental logic

The experiment is organized by theorem mechanism rather than by a collection of generic posterior models.

Claimed scale	Numerical construction	Diagnostic
$T_{\mathrm{glob}} = \Theta(\kappa_\star)$	2-D logarithmic spiral	fixed mean contraction time
$N_{\mathrm{glob}} = \Theta(\kappa^2)$	1-D flat-top bump train	fixed energy-gap reduction
$\gamma^{-1} = \Theta(\log\kappa)$	Chen et al. smoothed bump	smallest local eigenvalue
$\gamma^{-1} = \Theta(\sqrt{n\log\kappa})$	gamma shell	reduced full-operator gap
$\gamma^{-1} = \Theta(\sqrt{\kappa})$	convex ridge, gamma shell	reduced full-operator gap
local flow / two discrete maps	ridge and shell	nonlinear/linear contraction ratio
global-to-local transition	bump train plus quadratic core	entry and tail complexity

The logarithmic one-dimensional construction follows Theorem 5.7 of Chen et al.; all other constructions follow the sharpness appendices and the attached local-rate note. The diagnostics use the same quantities as the theorems: flow time for the continuous dynamics, iteration count for discrete dynamics, the Fisher–Rao tangent norm for local convergence, and the reverse-KL energy gap for the global bump-train experiment.

2 Numerical implementation

No ambient-dimensional matrices. The ridge and shell are $O(m)$-invariant in their transverse coordinates. Their full linearized operator decomposes into a three-dimensional scalar block, a two-dimensional vector block (multiplicity m), and a one-dimensional traceless block. We diagonalize

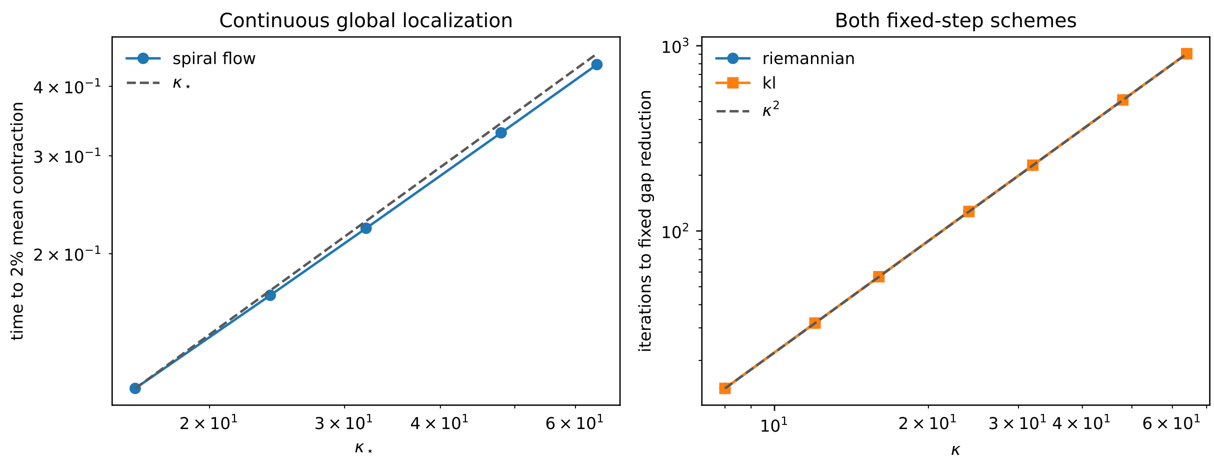

Figure 1: Global sharpness. The logarithmic-spiral flow has a fixed-contraction time proportional to $\kappa_\star$. At the globally safe fixed step $\Delta t = 0.5/\kappa$, both covariance maps require order κ^2 iterations for a fixed energy reduction.

all three blocks, so the reported γ is not merely the Rayleigh quotient of the analytical witness. Gaussian expectations reduce to (T, U) with $T \sim \mathcal{N}(0,1)$ and $U \sim \chi^2_m$. Gauss–Hermite quadrature is used for T and a composite Gauss–Legendre rule in chi-square survival-probability coordinates for U. This permits ridge dimensions as large as $n = K^4/8 + 1 = 33{,}554{,}433$ at $K = 128$.

Independent nonlinear check. For representative shells and ridges, the full nonlinear dynamics remain in the reduced state (m_t, c_t, c_y). We initialize along the slow eigenvector at amplitude 2×10^{-5} and integrate the exact reduced expectations. For the Riemannian scheme we use $\Delta t = 0.5/\Lambda$; for KL/Bregman we use the same step and its exact stepsize-weighted tangent norm. Finite-difference tests independently compare the nonlinear Jacobian with the Bochner-form operator.

Global constructions. The spiral flow is integrated with tensor Gauss–Hermite expectations in two dimensions. Every accepted state is checked to keep every quadrature point in the flat spiral core. The bump train uses a C^∞ flat-top profile, $\Delta t = 0.5/\kappa$, and exact one-dimensional Gauss–Hermite expectations. Disjoint supports allow saturated bumps to be accumulated analytically. Both schemes shadow the prescribed centers to floating-point precision and keep $\kappa c_n = 1$ throughout the train.

3 Results

The continuous fit over $K \in \{16, 24, 32, 48, 64\}$ is

$$T_{0.98} \propto \kappa_\star^{0.967955}, \qquad R^2 = 0.999957.$$

The largest normalized phase displacement of any accepted quadrature point is below one in every run, so the simulation remains inside the formula's flat core. For both discrete methods, over $\kappa \in \{8, 12, 16, 24, 32, 48, 64\}$,

$$N_{0.8} \propto \kappa^{2.00091}, \qquad R^2 > 0.999999.$$

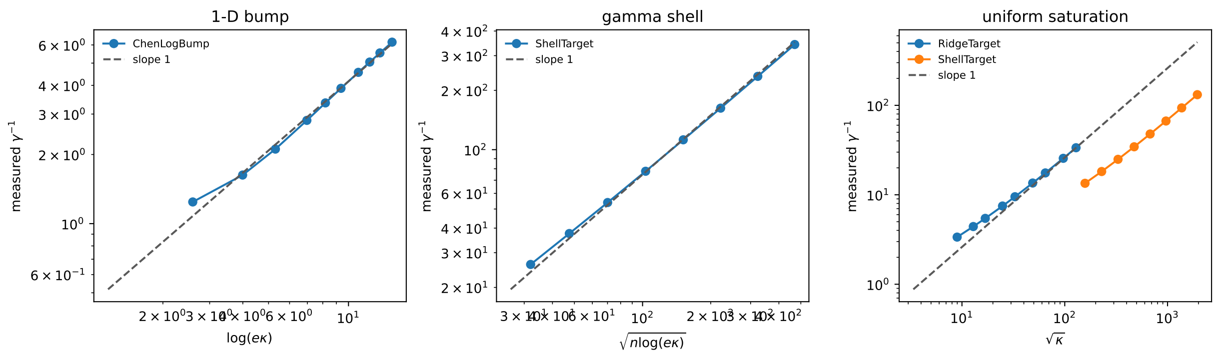

Figure 2: Local sharpness for the three mechanisms. Each horizontal coordinate is the rate scale predicted for that family; slope one is the sharpness prediction. The saturation panel contains two independent constructions.

Table 1: Log–log exponent of measured γ^{-1} against the predicted scale. The sharp prediction is one.

Mechanism	all points	asymptotic half
one-dimensional $\log(e\kappa)$	0.9645	1.0448
gamma-shell $\sqrt{n\log(e\kappa)}$	0.9577	0.9826
ridge $\sqrt{\kappa}$ saturation	0.8674	0.9229
shell $\sqrt{\kappa}$ saturation	0.9097	0.9570

The equality of the two curves is intentional: on the bump centers $c_n A(m_n, c_n) = 1$ to numerical precision, and both covariance maps fix $c_n = 1/\kappa$.

Table 1 reports both the full-grid exponent and the upper-half tail exponent. The latter is the more relevant asymptotic diagnostic because the constructions contain visible additive constants at small K, m. The normalized ratios $\gamma^{-1}/s_{\text{predicted}}$ remain bounded and approach nonzero constants. The ridge simultaneously retains the analytical Hessian-Lipschitz certificate $L_H \le 64$.

At $\kappa = 16$, local entry occurs after 751 Riemannian iterations and 753 KL/Bregman iterations. Continuing at the global step reaches norm 10^{-6} in 1,113 and 1,115 total iterations, respectively; the two-stage versions need only 768 and 770. Thus the stepsize switch removes 345 tail iterations for each method. This is the numerical global-to-local decomposition in the end-to-end theorem: the common global prefix is unchanged, while the local tail benefits from the larger locally admissible step.

4 Scope and non-claims

The experiment preserves the qualifications of the lower bounds.

- The spiral uses the paper's admissible anisotropic spectral initialization; it is not evidence for a κ lower bound from $C_0 = \beta^{-1} I$.
- The discrete κ^2 result is for a globally fixed $\Delta t = \Theta(\kappa^{-1})$ and a constant-factor energy reduction. It does not lower-bound line searches, adaptive curvature steps, or the complete $\log(\Delta_0/\delta)$ accuracy factor.
- The gamma shell establishes dimensional sharpness under the curvature box; its Hessian-Lipschitz modulus is not dimension-uniform. Uniform Hessian-Lipschitz saturation is supplied

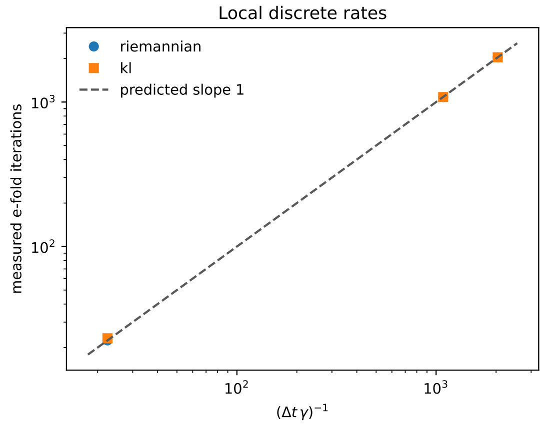

Figure 3: Measured e-fold iterations of both local schemes against $(\Delta t\,\gamma)^{-1}$. The nonlinear iterations agree with their exact linearized maps; KL/Bregman uses its stepsize-weighted metric.

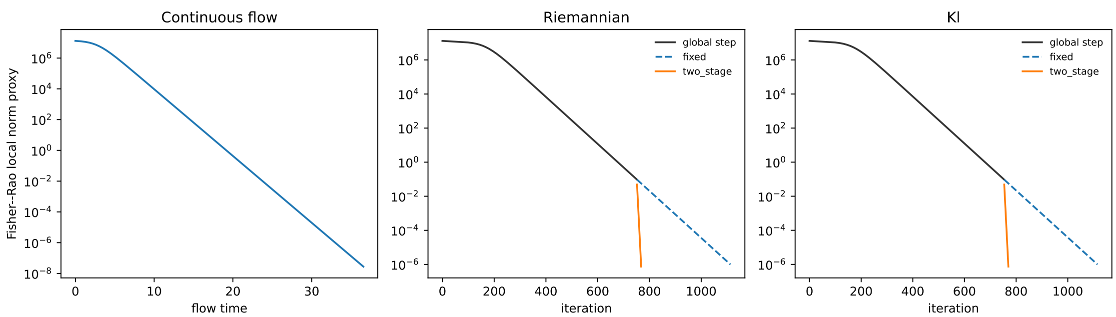

Figure 4: A single bump-train target exhibits a long far-field phase followed by unit-scale quadratic local contraction. For both discrete schemes, the two-stage run shares the entire globally safe prefix with the fixed-step run, then switches from $0.5/\kappa$ to 0.5 after local entry.

separately by the ridge.

- The near-saturation strip identified as open in the local-rate note is not claimed to be resolved numerically.

5 Reproduction

From the repository root:

```
python scripts/natural_gradient_sharpness/run_experiments.py \
  --config configs/natural_gradient_sharpness/sharp_rates.yaml \
  --outdir outputs/natural_gradient_sharpness --overwrite

python scripts/natural_gradient_sharpness/plot_results.py \
  --outdir outputs/natural_gradient_sharpness

pytest -q tests/natural_gradient_sharpness/test_constructions.py
```